

Amit Kumar Yadav

Senior Software Engineer

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Professional Summary

Senior Software Engineer with **8.5+ years** of experience designing scalable, cloud-native, event-driven distributed systems for enterprise clients, actively targeting **Software Architect** roles. Expert in microservices architecture (Java 25, Spring Boot, Kafka), polyglot persistence (Neo4j, PostgreSQL), container orchestration (Docker/Kubernetes), and full-stack observability. Hands-on practitioner of **Generative AI** — built intelligent data retrieval services using **RAG pipelines and vector embeddings**, improving contextual search accuracy — with an M.Tech in Data Science from BITS Pilani (CGPA 8.2).

Work Experience

Senior Software Engineer — Siemens Technology and Services 11/2021 – Present

Distributed Systems Engineering | Cloud-Native Microservices | Platform Architecture Bangalore, India

- Owned and led the design & development of **5+ microservices** in an **event-driven architecture** using **Java 25, Spring Boot, and Spring API Gateway**, supporting a multi-tenant SaaS platform that leveraged **Apache Kafka, Neo4j, and PostgreSQL**.
- Designed and optimized API and persistence layers, leveraging **Neo4j graph queries** for complex relationship intelligence and **PostgreSQL with Flyway migrations**, resulting in **30–40% improvement in query performance** and better data consistency.
- Deployed containerized applications using **Docker and Kubernetes**, achieving **99.9% service availability**, reducing deployment time by **50%**, and enabling seamless **CI/CD pipelines**.
- Developed and maintained comprehensive API documentation using **Swagger / OpenAPI**, improving developer onboarding time by **40%** and ensuring consistent, **well-defined API contracts** across services.
- Implemented scalable **multi-tenancy architecture**, enabling onboarding of multiple enterprise clients, improving system scalability by **3x**, and ensuring strict **tenant isolation**.
- Established full-stack observability using **ELK Stack, Datadog, and Grafana**, reducing mean time to resolution (MTTR) by **60%** through proactive monitoring and real-time alerting.
- Knowledge on **GenAI-powered** intelligent data retrieval services, leveraging **RAG pipelines and vector embeddings**.

Software Engineer — Tarento Technology 07/2020 – 11/2021

iGOT — Igot karmayogi platform | Govt. of India Bangalore, India

- Developed the **workflow microservice from scratch** using Java 8 and Spring Boot, handling content lifecycle and validation for **10,000+** government training records on a platform serving **millions of users**.
- Built content and validation services with Elasticsearch and PostgreSQL, reducing query latency by **35%** and achieving **95%+ uptime**, cutting downtime incidents by **50%** under national-scale concurrent load.
- Owned end-to-end backend delivery — development, unit testing, debugging, and production issue resolution — designing scalable microservices on Cassandra to sustain high government user traffic.

Software Engineer — Tarento Technology 10/2019 – 07/2020

Water & Sewerage Municipal Management | eGovernments Foundation Bangalore, India

- Developed and owned end-to-end setup of the water & sewerage tax and billing system for **5+ municipal corporations**, handling project setup, development, and ongoing maintenance.
- Integrated the module with the broader municipal microservices ecosystem using Spring Boot, Apache Kafka, and Spring Data JPA, enabling real-time event streaming for **100K+ transactions/month** on MySQL.
- Resolved critical performance bottlenecks through targeted debugging, delivering a stable and maintainable platform with zero post-launch critical defects.

Software Engineer — Zwayam Digital 09/2017 – 09/2019

RBAC | Workforce Management (WFM) — Client: Mindtree Bangalore, India

- Designed and developed the **RBAC system from scratch** (Java 8, Spring Boot, MongoDB), implementing secure role and permission management and achieving **100%** elimination of unauthorized-access incidents post-deployment.
- Built the WFM system to optimize workforce productivity and allocation, contributing strategic feature enhancements that reduced manual HR workflow steps by **30%** across enterprise clients.
- Led end-to-end design, development, and client support (Mindtree) for two concurrent modules with AngularJS front-end integration, delivering both on time with **zero client-reported defects**; awarded **First Place Project Award** at the 2019 annual review.

Education

M.Tech — Data Science (WILP)	2023 – 2025
BITS Pilani	CGPA: 8.2 / 10
B.Tech (Honours) — Computer Science & Engineering	07/2013 – 05/2017
Institute of Engineering and Rural Technology, Prayagraj	Aggregate: 78%

Technical Skills

Languages: Java 25, Python
Architecture: Microservices, Event-Driven Architecture, Distributed Systems, Domain-Driven Design, API Gateway Design
Frameworks: Spring Boot, Spring API Gateway, Spring Microservices, Spring Data JPA, Hibernate
Messaging: Apache Kafka
Databases: PostgreSQL, Neo4j (Graph), MongoDB, Apache Cassandra, MySQL
DevOps / Observability: Docker, Kubernetes, Flyway, ELK Stack, Datadog, Grafana
AI / ML: Generative AI, RAG Pipelines, Vector Embeddings, LLM Integration, Data Science
Testing: JUnit, Mockito
Tools: Git, REST APIs, JSON

Certifications & Training

Software Designing and Architecture Training (SDAT)	2024
Enterprise software architecture, design patterns, system design at scale, and solution architecture governance.	

Awards & Recognition

- Annual Segment Meeting Award — Siemens
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- Limelite Award — Tarento Technology, Q3 2020
- First Place Project Award — Zwayam Digital, Q4 2019